INTERNATIONAL SEARCH REPORT

International application No
PCT/US2021/037415

A. CLASSIFICATION OF SUBJECT MATTER
INV. G01N33/569 A61K49/00 C12Q1/04 A61B5/145 A61L15/36
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
G01N A61K C12Q A61B A61L

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)
EPO-Internal, WPI Data, EMBASE, BIOSIS

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	WO 2019/209982 A1 (LABORATORY CORP AMERICA HOLDINGS [US]) 31 October 2019 (2019-10-31) para 9, 89, 114-115, 150, 154 ; claim 16, 33, 36 ; example 1	1,3-14, 16,18
X	CA 2 993 336 A1 (ENBIOTIX INC [US]) 26 January 2017 (2017-01-26)	1-3,10, 16-18
Y	claims 1, 9-10, 28-29, 48-49 ; para 2-3	2-16
X	US 8 182 804 B1 (COLLINS JAMES J [US] ET AL) 22 May 2012 (2012-05-22)	17,18
Y	example 1 ; col 3, l 5-25 ; col 17, l 5-22	2-16
X	US 2012/143024 A1 (PHILLIPS ERICA M [US] ET AL) 7 June 2012 (2012-06-07)	1,18
Y	para 34 ; claims	2-16

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[X] Further documents are listed in the continuation of Box C.

[X] See patent family annex.

* Special categories of cited documents :

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"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

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Date of the actual completion of the international search	Date of mailing of the international search report
1 September 2021	09/09/2021

Name and mailing address of the ISA/	Authorized officer
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International application No
PCT/US2021/037415

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	US 9 457 132 B2 (DONLAN RODNEY MARTIN [US]; LEHMAN SUSAN MARIE [US] ET AL.) 4 October 2016 (2016-10-04) examples -----	1-18

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INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No
PCT/US2021/037415

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
WO 2019209982 A1	31-10-2019	AU 2019261608 A1	15-10-2020
		BR 112020020622 A2	12-01-2021
		CA 3096281 A1	31-10-2019
		CN 112313337 A	02-02-2021
		EP 3784786 A1	03-03-2021
		US 2021079443 A1	18-03-2021
		WO 2019209982 A1	31-10-2019
CA 2993336 A1	26-01-2017	CA 2993336 A1	26-01-2017
		EP 3324987 A1	30-05-2018
		JP 2018529755 A	11-10-2018
		US 2019167736 A1	06-06-2019
		WO 2017015652 A1	26-01-2017
US 8182804 B1	22-05-2012	US 8182804 B1	22-05-2012
		WO 2006137847 A2	28-12-2006
US 2012143024 A1	07-06-2012	AU 2008334326 A1	18-06-2009
		CA 2705837 A1	18-06-2009
		EP 2229592 A2	22-09-2010
		JP 5956110 B2	20-07-2016
		JP 5993048 B2	14-09-2016
		JP 2011507494 A	10-03-2011
		JP 2015107130 A	11-06-2015
		US 2012143024 A1	07-06-2012
		WO 2009074893 A2	18-06-2009
US 9457132 B2	04-10-2016	CA 2837731 A1	04-04-2013
		US 2014199360 A1	17-07-2014
		WO 2013048604 A2	04-04-2013